

Table C.25: Model Selection: Abnormal without Accounting for Measurement Error

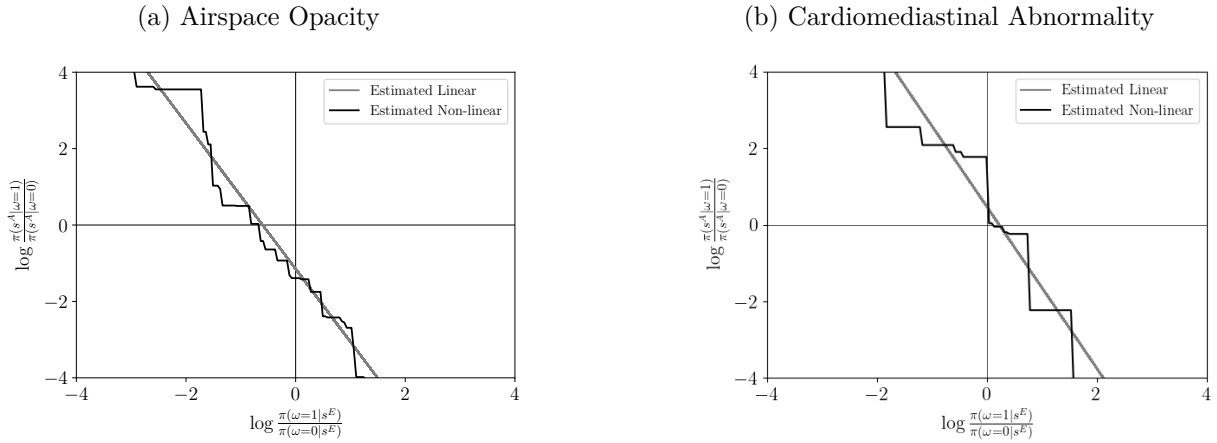
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
AUTOMATION BIAS (b)	0.47 (0.02)	0.35 (0.02)	0.44 (0.03)	0.42 (0.02)	0.37 (0.02)	0.37 (0.02)	0.36 (0.02)	0.36 (0.02)	0.36 (0.02)	0.41 (0.02)	0.37 (0.02)	0.36 (0.02)
OWN INFO BIAS (d)	0.92 (0.02)	0.89 (0.02)	0.86 (0.02)	0.90 (0.02)	0.88 (0.02)	0.88 (0.02)	0.85 (0.02)	0.88 (0.02)	0.82 (0.02)	0.87 (0.02)	0.87 (0.02)	0.88 (0.02)
CONSTANT	1.23 (0.07)	0.98 (0.06)	1.16 (0.08)	1.12 (0.07)	1.02 (0.06)	1.05 (0.06)	0.99 (0.06)	1.04 (0.06)	1.01 (0.07)	1.11 (0.07)	1.05 (0.07)	1.00 (0.06)
FOCAL s_A	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
OTHER s_A		✓			✓	✓		✓			✓	✓
FOCAL s_E			✓	✓	✓	✓			✓	✓	✓	✓
OTHER s_E				✓		✓				✓		✓
CLINICAL HISTORY s_E							✓	✓	✓	✓	✓	✓
J-STATISTIC	12.09	7.1	0.35	4.71	3.06	4.94	0.0	5.79	0.0	0.34	0.0	4.85
MMSC-BIC	-13.23	-5.56	-3.87	-3.73	-1.16	-3.5	0.0	-2.65	0.0	-3.88	0.0	-3.59
NUMBER OF MOMENTS	9	6	4	5	4	5	3	5	3	4	3	5
Observations	5710	5710	5710	5710	5710	5710	5710	5710	5710	5710	5710	5710
R-SQUARED	0.56	0.54	0.53	0.54	0.54	0.54	0.54	0.54	0.53	0.53	0.54	0.54

Note: This table presents results of the model selection exercise as described in Table 3 for abnormal without accounting for measurement error in the radiologist reports.

C.5.5 Linearity of the Update Model

To assess the appropriateness of linear relationship in the model of radiologist updating with AI we estimate non-parametric versions of the model and plot the empirical analog of Figure 5 along with the joint distribution of signals. To do so, we estimate a boosted tree that estimates the radiologist's reported $\frac{p_r(\omega_i=1|s_{ir}^A, s_{ir}^E)}{p_r(\omega_i=0|s_{ir}^A, s_{ir}^E)}$ as a non-parametric function of a constant, the update term, and the reported probability without AI assistance. We impose monotonicity constraints on the update term and the reported probability without AI. We then plot the frontier in which radiologists are indifferent between following up on the case-pathology and not following up. We compare this frontier to the cutoff frontier of a Bayesian decision maker, the radiologist without AI assistance, and the radiologist with AI assistance under the linear model.

Figure C.38: Empirical analog of figure 5



Note: For the two top-level pathologies with AI assistance, we plot estimates of the indifference frontier where radiologists are indifferent between following up on a patient-case and not following up on a patient-case for a Bayesian decision maker, the linear model estimated in equation 7, a non-parametric version of equation 7, and the human only without AI assistance. In addition, we plot the joint distribution of the signals log-likelihoods.

C.5.6 Individual Heterogeneity

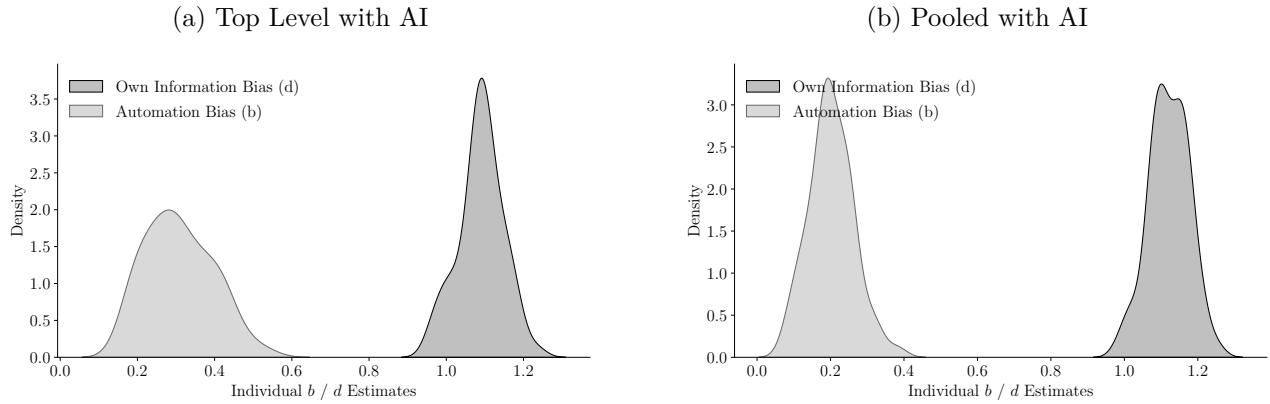
Here we show the distribution of individual estimates of equation (7). Each estimate contains sampling error, as each radiologist is only reading a subset of cases. Therefore, the unadjusted distribution of individual-level estimates is overdispersed as it is a convolution of the true individual-level parameters and the sampling noise. We adjust for this over-dispersion using

a Bayesian hierarchical model, where we model the individual parameter vector θ_r as follows.

$$\begin{aligned}\theta_r &\sim N(\mu, \Sigma) \\ \mu &\sim N(0, 100I) \\ \Sigma &= \text{diag}(\tau) \Omega \text{diag}(\tau) \\ \tau_k &\sim \text{Cauchy}(0, 2.5) \\ \Omega &\sim \text{LKJCorr}(10)\end{aligned}$$

We sample from the posterior of this model and plot the marginal distribution of the posterior means of θ_r below.

Figure C.39: Individual Heterogeneity in b and d



Note: Marginal distributions of individual b and d estimates by radiologist for top level pathologies with AI and all pathologies with AI.

C.6 Preference Estimation

In the experiment we elicit both probability assessment and treatment decisions, allowing us to identify the relative costs of false positives and false negatives the radiologists are using. Recall that radiologist r chooses to treat or follow-up on pathology p in patient case i under treatment t if $a_{ritp} = 1$ where a_{ritp} is given by

$$a_{ritp} = 1 \left[\frac{p_{ritp}}{1 - p_{ritp}} - c_{rel}^{rp} + \varepsilon_{ritp} > 0 \right]$$

where p_{ritp} is the radiologist's probability assessment, c_{rel}^{rp} is the relative cost of false positives and false negatives for radiologist r and pathology p , and ε_{ritp} captures unobserved preference heterogeneity. If ε_{ritp} follows a Logistic distribution, we can estimate c_{rel}^{rp} through a logistic

regression. We impose a low-dimensional structure on c_{rel}^{rp} to improve statistical precision and estimate the following logistic regression

$$\log \frac{P(a_{ritp} = 1)}{1 - P(a_{ritp} = 1)} = \beta_0 + \beta \log \frac{p_{ritp}}{1 - p_{ritp}} + \alpha_p + \gamma_r \quad (9)$$

where α_p are pathology fixed effects and γ_r are radiologist fixed effects. The relative costs of false positives to false negatives for radiologist r and pathology p can then be found as $c_{rel}^{rp} = \exp \left[-\frac{\beta_0 + \gamma_r + \alpha_p}{\beta} \right]$. For each pathology, we winsorize radiologists' relative costs at the 5th and 95th percentile. The results of this exercise are presented in Table C.26.

Table C.26: Preference Estimates

	Mean	Std	Min	25%	50%	75%	Max
All	3.926	5.754	0.233	0.655	1.521	4.040	22.336
Top (with AI)	1.228	1.669	0.149	0.266	0.482	1.259	6.362
AI	2.261	3.312	0.184	0.379	0.867	2.307	13.045
Abnormal	2.459	3.326	0.344	0.537	1.016	2.551	12.604

Note: Distribution of c_{rel}^{rp} for each of the four pre-registered pathology groups calculated from the estimates of equation 9. The distribution of c_{rel}^{rp} is winsorized for each pathology at the 5th and 95th percentile.